KENYATTA UNIVERSITY

DEPARTMENT OF MATHEMATICS AND ACTURIAL SCIENCE.

BACHELOR OF SCIENCE(STATISTICS AND PROGRAMMING), BACHELOR OF SCIENCE (GENERAL),BACHELOR OF EDUCATION,BACHELOR OF ARTS.

SST408 DESIGN AND ANALYSIS OF SAMPLE SURVEYS II.

CAT1 23rd Feb. 2021 2-4pm.

ATTEMPT ALL QUESTIONS.

QUESTION ONE(15 MARKS).

a) Draw a simple random sample of size n^1 from a population of size N and estimate $\bar{y}_{s^1}$. Draw another simple random sample of size n from n^1 and estimate $\bar{y}_s$. Prove that

i) $Cov(\bar{y}_s, \bar{y}_{s^1}) = Var(\bar{y}_{s^1})$.

ii) $Cov(\bar{y}_{s^1}, \bar{y}_s - \bar{y}_{s^1}) = 0$.

iii) $Var(\bar{y}_{s^1} - \bar{y}_s) \geq 0$. (6 marks)

b) A survey was conducted in a district in two consecutive years to estimate the total pig population. The district has 77 villages. A sample 12 villages was selected with simple random sampling in the first year. In the second year the survey was confined to a subsample of 6 villages selected from the 12 villages. The data collected was as follows;

S.N of Village	S_x^2	X Pig population First Year	y_i Second Year	S_y^2
1	709.8	220	189	402.5^2
2	407.2	1337	758	166.5^2
3	275.8	654	583	8.5^2
4	256.2	1186	681	89.5^2
5	853.2	1783	1195	603.5^2
6	530.8	399	143	448.5^2
7		2212		

N = 77
n' = 12 – first stage
n = 6 – second stage

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Q1, 2, 3.

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (STATISTICS AND PROGRAMMING GENERAL) BACHELOR OF EDUCATION AND BACHELOR OF ARTS

SST 408: DESIGN AND ANALYSIS OF SAMPLE SURVEYS II

DATE: Tuesday, 18th May 2021 TIME: 8.00a.m-10.00a.m

INSTRUCTIONS:

Answer Question ONE and any other TWO questions.

QUESTION ONE(30 MARKS).

a) Distinguish clearly the difference between single stage cluster sampling and two stage cluster sampling. (4 marks)

b) If $\hat{\theta}$ denotes an estimator of a population parameter θ, Prove that under double phase sampling.

i) $E(\hat{\theta}) = E_1 E_2(\hat{\theta})$ (2 marks)

ii) $V(\hat{\theta}) = V_1[E_2(\hat{\theta})] + E_1[V_2(\hat{\theta})]$ (5 marks)

c) A set of 20,000 records are stored in 400 file drawers each containing 50 records. Five records are drawn at random from each of the 8 randomly selected drawers;

(i) Identify which sampling design is used to select the sample. (2 mark)

(ii) If on one item the estimates of the variances were $s_b^2 = 362$ and $s_w^2 = 805$

Calculate the sampling error of the estimator of the population mean. (4 marks)

d) In a survey we want to improve the precision of estimating the population mean using two estimators, the ratio and regression estimator. However the value of the design variable is unknown.

i) Describe the sampling design which can be used and explain how we can now use the two estimators. (3 marks)

ii) If the size of the samples $s_1 = 300$ and $s_2 = 87$ such that $s_2 \subset s_1$ assuming N is very large and given the following computations;

$$\sum_{i=1}^{n}(y_i - \bar{y}_s)^2 = 17283 \;,\; \sum_{i=1}^{n}(x_i - \bar{x}_s)^2 = 3248 \text{ and } \sum_{i=1}^{n} x_i - \bar{x}_s)(y_i - \bar{y}_s) = 5114.$$

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And $\hat{R} = \frac{\bar{y}}{\bar{x}} = 2.3$

Calculate the estimated variances of the estimator of the population mean using the ratio estimator with double phase sampling and the simple random sampling design. Determine which of the two estimators is more precise. (8 marks)

e) What is imputation?Differentiate between cold deck and Hot deck imputation. (2 marks)

QUESTION TWO(20 MARKS)

a) Define the ratio estimator of the population mean. Explain under which circumstances we use double phase sampling with the ratio estimator. (2 marks)

b) Prove that the variance of the ratio estimator under double phase sampling is given by ;

$$V(\bar{y}_R) = \left(\frac{1}{n^1} - \frac{1}{N}\right)S_Y^2 + \left(\frac{1}{n} - \frac{1}{n^1}\right)(S_Y^2 + R^2 S_X^2 - 2RS_{YX})$$

Given the following expectations.

$$E(e_2^2) = \left(\frac{1}{n^1} - \frac{1}{N}\right)\frac{1}{\bar{X}^2}S_x^2$$

$$E(ee_1) = \left(\frac{1}{n} - \frac{1}{N}\right)\frac{S_{xy}}{\bar{X}\bar{Y}} = \left(\frac{1}{n} - \frac{1}{N}\right)C_{xy} \text{ and } E(e_1^2) = \left(\frac{1}{n} - \frac{1}{N}\right)\frac{S_x^2}{\bar{X}^2}$$

c) A survey was conducted in a district in two consecutive years to estimator the average number of cows .The district has 40 villages. A sample of 10 villages were selected with equal probability without replacement in the first year. In the second year the survey was confined to a subsample of 5 villages selected from the 10 villages. The number of cows in the selected villages are given below. (10 marks)

S.N of Village	Total Number of Cows. First Year	Second Year
1	220	189
2	654	583
3	228	111
4	813	616
5	528	267
6	237	
7	681	
8	502	
9	405	
10	542	

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Using double phase sampling with ratio estimator, estimate the;

i) Total pig population and its standard error. (6 marks)

ii) Variance using simple random sampling and compare the efficiency of the two designs. (2marks)

QUESTION THREE(20 MARKS)

a) Describe how you select a sample using cluster sampling design. (2 marks)

b) If $\bar{\bar{y}}$ is the cluster sampling mean estimator of the population mean, Prove that

$$V(\bar{\bar{y}}) = \frac{(N-n)}{Nn} S_b^2$$

Where $S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{Y}_i - \bar{\bar{Y}})^2$ n, N denotes the number of clusters in sample and population. (8 marks)

c) A survey on the amount of purple tea produced in 45 tea producing districts which form our clusters each with 12 villages was conducted to estimate the average amount of purple tea produced in the country. For this purpose ,3 districts were selected from 45 districts by simple random sampling without replacement. The information on the amount of purple tea produced in the selected villages is given below;

Cluster Number	Cluster size	Yields of purple tea produced in each village
1	12	86,85,57,81,71,92,72,37,51,81,50,43
2	12	86,45,81,55,56,55,91,70,64,19,62,41
3	12	81,43,67,48,47,35,34,37,78,71,79,80

Using the two stage sampling design estimate;

i) Mean yield per acre with its standard error. (5 marks)

ii) Variance using simple random sampling and compare the efficiencies of the two designs. (5 marks)

QUESTION FOUR.(20 MARKS)

(a) Describe Rotation sampling design and explain under which conditions it reduce to;

i) Panel design.

ii) Cross sectional design. (5 marks)

(b) In rotation design if on the second occasion the' n-m' units are selected from N units the variance of estimate of change $\hat{\Delta}$ where $\hat{\Delta} = \hat{Y}_2 - \hat{Y}_1$ is given by

$$V(\hat{\Delta}_1) = (1-\frac{n}{N})\frac{S_1^2}{n} + (1-\frac{m^2+(n-m)^2}{nN})\frac{S_2^2}{n} - \frac{(1-n/N)}{n}\rho S_1 S_2$$

(8 marks)

(c) A simple random sample of 10 cities was selected from 100 cities in 1980-81 for estimating the total number of inhabitants. Of these 5 cities were retained which constitute the matched sample for the observation in 1981-82 and a fresh sample(unmatched) of 5 cities was selected from the remaining 95 cities independently. This data is given below;

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S.No of Cities	Number of Inhabitants. 1980-81	1981-82	S.No	1981-82
1	40	64	11	80
2	40	60	12	143
3	56	142	13	67
4	64	77	14	115
5	77	89	15	69
6	76			
7	138			
8	67			
9	29			
10	23			

Estimate the change in the number of inhabitants in 1981-82 as compared to 1980-81 along with its standard error. (7 marks)

QUESTION FIVE (20 MARKS).

a) In rotation design a matched sample can be taken as a double phase sample. In phase one we take a simple random sample of size n. In phase two we have two samples matched (m) and unmatched (m^1) .To get an estimator of $\bar{y}_2$ using both the matched and unmatched samples we use a compromise rotation estimator

$$\bar{y}_2^* = \phi \bar{y}_{2m} + (1-\phi)\bar{y}_{2m^1} \text{ where } \phi \text{ is unknown.}$$

Making assumptions about ϕ ,Prove that $\hat{\phi} = \dfrac{w_{2m^1}}{w_{2m} + w_{2m^1}}$ where $w_{2m^1} = \dfrac{n}{s_2^2}$ and

$$w_{2m^1} = \frac{n+m}{ns_2^2(1-\rho^2)+ms_2^2\rho^2}.$$

and find the optimum value of variance of $\bar{y}_2$. (10 marks)

b) In milk surveys ,the amount of yield was recorded in 15 randomly selected villages in a sub-county for two consecutive years 1980-1981 and 1981-1982 to study the change in the milk yield. The first 5 villages was common to the two years .

S.No of Village	Milk yield per year. 1980-81	1981-82	S.No of village	Milk yield per year 1980-1981	1981-1982
.1	954	933	9	607	
.2	400	406	10	75	
3	536	677	11		145
4	73	96	12		211
5	364	357	13		641
6	205		14		648
7	41		15		604
8	893				

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Estimate the average milk yield and its sampling error in the sub-county using the compromise estimator together with its standard error. (10 marks).

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BACHELOR OF SCIENCE(STATISTICS AND PROGRAMMING), BACHELOR OF SCIENCE (GENERAL),BACHELOR OF EDUCATION,BACHELOR OF ARTS.

SST408 DESIGN AND ANALYSIS OF SAMPLE SURVEYS II.

CAT1 23rd Feb. 2021 2-4pm.

ATTEMPT ALL QUESTIONS.

QUESTION ONE(15 MARKS).

a) Draw a simple random sample of size n' from a population of size N and estimate $\bar{y}_{s'}$. Draw another simple random sample of size n from n' and estimate $\bar{y}_s$. Prove that

i) $Cov(\bar{y}_s, \bar{y}_{s'}) = Var(\bar{y}_{s'})$.

ii) $Cov(\bar{y}_{s'}, \bar{y}_s - \bar{y}_{s'}) = 0$.

iii) $Var(\bar{y}_{s'} - \bar{y}_s) \geq 0$. (6 marks)

b) A survey was conducted in a district in two consecutive years to estimate the total pig population. The district has 77 villages. A sample 12 villages was selected with simple random sampling in the first year. In the second year the survey was confined to a subsample of 6 villages selected from the 12 villages. The data collected was as follows;

S.N of Village	Pig population . First Year	Second Year
1	220	189
2	1337	758
3	654	583
4	1186	681
5	1783	1195
6	399	143
7	2212	

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N = 77

n' = 12.

n = 6.

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UNIVERSITY EXAMINATIONS 2018/2019

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE (STATISTICS AND PROGRAMMING), BACHELOR OF SCIENCE (GENERAL), BACHELOR OF EDUCATION, AND BACHELOR OF ARTS

SST 408: DESIGN AND ANALYSIS OF SAMPLE SURVEYS II

DATE: MONDAY 26TH NOVEMBER 2018 TIME: 8.00 A.M. – 10.00 A.M.

$V(\hat\theta) = V_1[E_2(\hat\theta)] + E_1[V_2(\hat\theta)]$

$= V_1(\hat\theta') + E_1\left[\frac{1}{n} - \frac{1}{n'}\right]S'^2$

$= \frac{1}{n'} - \frac{1}{N} S^2 + \left(\frac{1}{n} - \frac{1}{n'}\right) E_1(S'^2)$

But $E(S'^2) = S^2$

$= \left(\frac{1}{n'} - \frac{1}{N}\right)S^2 + \left(\frac{1}{n} - \frac{1}{n'}\right)S^2$

$= \left(\frac{1}{n'} - \frac{1}{N} + \frac{1}{n} - \frac{1}{n'}\right)S^2 = \frac{1}{n} - \frac{1}{N} S^2 = V(\hat\theta)$

INSTRUCTIONS

Answer Question One and any other Two questions

QUESTION ONE(30 MARKS).

a) Distinguish clearly the deference between single stage cluster sampling and two stage cluster sampling. (4 marks)

If $\hat\theta$ denotes an estimator of a population parameter θ. Prove that under double phase sampling.

$E(\hat\theta) = E_1E_2(\hat\theta)$

$E_2(\hat\theta) = \theta'$

$E_1(\hat\theta') = \theta$

And $E(\hat\theta) = \theta$

Hence equal

i) $E(\hat\theta) = E_1E_2(\hat\theta)$

$E_1((E_2(\hat\theta'/n'))/N)$

But $E_2(\hat\theta'/n') = \theta'$

$E_1(\hat\theta'/N) = \theta$

ii) $V(\hat\theta) = V_1[E_2(\hat\theta)] + E_1[V_2(\hat\theta)]$ (5 marks)

b) A set of 20,000 (N) records are stored in 400 (n) file drawers each containing 50 (M) records. Five records are drawn at random from each of the 8 (m) randomly selected drawers:

(i) Identify which sampling design is used to select the sample. Multi stage (1mark)

(ii) If on one item the estimates of the variances were $s_1^2 = 362$ and $s_2^2 = 805$.

Calculate the sampling error of the estimator of the population mean. ($\hat\theta$) (3 marks)

$V(\hat\theta) = V_1((E_2(\hat\theta/m))/n) + E_1((V_2(\hat\theta/m))/n)$

c) In a survey we want to improve the precision of estimating the population mean using two estimators, the ratio and regression estimator. However the value of the design variable is unknown.

SINGLE STAGE	MULTI ~~TWO~~ STAGE CLUSTER SAMPLING
1. M does not change when taking a cluster sample.	1. From M units, m units are selected from each of the selected n clusters.
2. Occurs in 2 steps/stages	2. Occurs in 3 steps/stages.
3. Enumerate all the units in the selected clusters.	3. Enumerate m units

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i) Describe the sampling design which can be used and explain how we can now use the two estimators. (5 marks)

ii) If the size of the samples $s_2 = 300$ and $s_1 = 87$ such that $s_1 \subset s_2$ assuming N is very large and given the following computations:

$$\sum_{i=1}^{n}(y_i - \bar{y}_2)^2 = 17283, \quad \sum_{i=1}^{n}(x_i - \bar{x}_2)^2 = 3248 \text{ and } \sum_{i=1}^{n} x_i - \bar{x}_2)(y_i - \bar{y}_2) = 5114.$$

Calculate the sampling error of the estimator of the population mean and determine which of the two estimators is more precise. (8 marks)

d) What is imputation? Differentiate between Cold and hot deck imputation and give one example each. (5 marks)

QUESTION TWO(20 MARKS)

The ratio estimator of the pop mean is given by $\bar{y}_{ratio} = \frac{\bar{y}}{\bar{x}}\bar{X}$

a) Define the ratio estimator of the population mean. Explain under which circumstances we use double phase sampling with the ratio estimator. (5 marks)

d ratio used when the auxiliary variable $\bar{X}$ is not known.

b) Prove that the variance of the ratio estimator under double phase sampling is given by:

$$V(\bar{y}_R) = \left(\frac{1}{n'} - \frac{1}{N}\right)S_Y^2 + \left(\frac{1}{n} - \frac{1}{n'}\right)(S_Y^2 + R^2 S_X^2 - 2RS_{YX})$$

(7marks)

c) A survey was conducted in a district in two consecutive years to estimator the average number of cows .The district has 40 villages. A sample of 10 villages were selected with equal probability without replacement in the first year. In the second year the survey was confined to a subsample of 5 villages selected from the 10 villages. The number of cows in the selected villages are given below.

S.N of Village	Total Number of Cows. First Year	Second Year
1	220	189
2	654	583
3	228	110
4	813	616
5	528	267.
6	237	
7	681	
8	502	
9	405	
10	542	

Using double phase sampling with ratio estimator, estimate the;

i) Total pig population and its standard error. (5 marks)

ii) Variance using simple random sampling and compare the efficiency of the two designs. (5 marks)

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(ii) Prove that the variance of this estimator $Var(\bar{y}) = \frac{(N-n)}{N}\frac{s_1^2}{n} + \frac{(M-m)}{m}\frac{s_2^2}{nm}$

Where $s_1^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \bar{Y})^2$ and $s_2^2 = \frac{1}{N-1}\sum_{i=1}^{N}\sum_{j=1}^{M}(Y_{ij} - \bar{Y}_i)^2$ (6 marks)

(c) At an experimental station, there were 100 fields shown with wheat. Each field was divided into 16 plots of equal size. Out of the 100 fields 10 fields were selected by simple random sampling without replacement. From each of the selected field 4 plots were selected by simple random sampling. The yiels are as follows;

Selected Field	Plots 1	2	3	4
1	4.32	4.84	3.96	4.04
2	4.16	4.36	3.50	5.00
3	3.06	4.24	4.76	3.12
4	4.00	4.84	4.32	3.72
5	4.12	4.68	3.46	4.02
6	4.08	3.96	3.42	3.08
7	5.16	4.24	4.96	3.84
8	4.40	4.72	4.04	3.98
9	4.20	4.66	3.64	5.00
10	4.28	4.36	3.00	3.50

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Using two stage sampling with , estimate the

(iii) Mean yield per acre with its standard error. (5 marks)

(iv) Variance using simple random sampling and compare the efficiency of the two designs.(3 marks)

QUESTION FOUR(20 MARKS)

(a) In an expenditure survey it was believed that there may be biased due to non response bias, Describe the various sources of non response and suggest procedures correcting this bias. (12 Marks)

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(b) Imputation is a common technique for handling non response in a survey . Prove that the variance of the mean when imputation is done is given by;

$$V(\bar{y}) = V_{sm} + V_{im,p} + 2V_{mix}$$

Where

$V_{im,p} = E_p(\bar{y}_I - \bar{y})^2 / s$ imputation variance

$V_{mix} = E_p E_s(\bar{y}_I - \bar{y})(\bar{y} - \bar{Y}) / s$ Mixed term

$V_{sm,p} = (\frac{1}{n} - \frac{1}{N})s^2$ sampling variance. (8 marks)

QUESTION FIVE (20 MARKS)

(a) Differentiate between panel and Rotation design. (4 marks)

(b) In milk surveys one may be interested in estimating change in the annual yield by using

(i) $\bar{y}_T - \bar{y}_{T-1}$ or

(ii) $\bar{y}_T + \bar{y}_{T-1}$ Identify the two designs and suggest which of the two is better and why.

(c) Show that the best linear estimator for the above two designs is a composite design

$\bar{y}^*_2 = \phi \bar{y}_{2m(reg)} + (1-\phi)\bar{y}_{2m^1}$ where

$\bar{y}_{2m(reg)} = \bar{y}_{2m} + b((\bar{y}_1 - \bar{y}_{1m})$ and $\bar{y}_{m^1} = \frac{1}{m^1}\sum_{i=1}^{m^1} \bar{y}_{hi}$

Prove that optimum value of $V(\bar{y}^*_2) = \frac{1}{w_{2m} + w_{2m^1}}$ where $w_{2m^1} = \frac{m^1}{s_2^2}$ and

$$w_{2m^1} = \frac{mn}{ms_2^2\rho^2 + ns_2^2(1-\rho^2)}$$

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(b) Impuation is a common technique for handling non response in a survey . Prove that the variance of the mean when imputation is done is given by;

$$V(\bar{y}) = V_{smp} + V_{imp} + 2V_{MIX}$$

Where

$V_{imp} = E_p(\bar{y}_i - \bar{y})^2 / s$ imputation variance

$V_{MIX} = E_p E_q(\bar{y}_i - \bar{y})(\bar{y} - \bar{Y}) / s$ Mixed term

$V_{imp} = (\frac{1}{n} - \frac{1}{N})s^2$ sampling variance. (8 marks)

QUESTION FIVE (20 MARKS)

(a) Differentiate between panel and Rotation design. (4 marks)

(b) In milk surveys one may be interested in estimating change in the annual yield by using

(i) $\bar{y}_T - \bar{y}_{T-1}$ or

(ii) $\bar{y}_T + \bar{y}_{T-1}$ Identify the two designs and suggest which of the two is better and why.

(c) Show that the best linear estimator for the above two designs is a composite design

$\bar{y}^*_2 = \phi \bar{y}_{2m(reg)} + (1-\phi)\bar{y}_{2m^1}$ where

$\bar{y}_{2m(reg)} = \bar{y}_{2m} + b(\bar{y}_1 - \bar{y}_{1m})$ and $\bar{y}_{m^1} = \frac{1}{m^1}\sum_{i=1}^{m^1} \bar{y}_{ki}$

Prove that optimum value of $V(\bar{y}^*_2) = \frac{1}{w_{2m} + w_{2m^1}}$ where $w_{2m^1} = \frac{m^1}{s_2^2}$ and

$$w_{2m^1} = \frac{mn}{ms_2^2\rho^2 + ns_2^2(1-\rho^2)}$$

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